AWG19 POLARIZATION DRIVER — CONNECTED SCHEMATIC

NUCLEO-F446RE: see NUCLEO_WIRING.md for cable pin numbers.

U3 LTC1693-5: POL, GND, GND, NC; IN, PHASE, GND, NC; VCC +12V, OUT DR4, NC, NC. R2 22. R5 100k to +12V. Q4 IRF9640 (G4).

U5 SN74AHCT125PWR: GND, POL_3V3, GND, TG_3V3, GND, +5V, GND, +5V; 10E, 1A, 20E, 2A, 3A, 30E, 4A, 4OE; 1Y POL, 2Y TG, 3Y, 4Y, VCC +5V, GND GND.

J3 NUCLEO_F446RE_CABLE: POL_3V3 1 PA8_/_POL; TG_3V3 2 PA9_/_TG; RX 3 PB10_/_RX; SH 4 PB4_/_SH; DP 5 PB5_/_DP; RESET 6 PA10_/_RESET; PA0_/_I_ADC 7 I_ADC; PC7_/_FF 8 FF; GND 9 GND; 3V3_FROM_MCU 10 +3V3; 5V_FROM_MCU 11 +5V; GND 12 GND.

R11 100k POL_3V3 GND. R12 100k TG_3V3 GND. R13 100k RX GND. R14 100k SH GND.

C9 100n / 16V +5V GND.

R15 100 DP GND. R16 100 RESET GND. R17 10k POL GND. R18 10k TG GND.

TOP. CLAMP_MID. D1 SMBJ28AD1. D2 US1D-E3/61T.

J2 EXTERNAL_AWG19: TOP 1, BOTTOM 2. Q5 IRF530 (G5). R6 100k to GND.

Q4 / Q5 need MCU dead time.
No hardware interlock yet.
Regulated 12 V supply required.

U5: 3.3V to 5V for gate drivers.
R11–R16: GPIO low when undriven. R17/R18: driver input pulldowns.
Power 3.3V and 5V from Nucleo only. J1–3 is a monitor/output.
No hardware interlock: firmware must prevent POL and TG overlap.

BOTTOM. U2A ADG5412F: BOTTOM 3, S, D 2 PRE. C1 1u. RX_IN.

J4 TO_AMPLIFIER: RX_IN 1, GND 2 GND.

U1 LTC1693-1: POL, TG, GND, GND; IN1, IN2, GND1, GND2; VCC1 +12V, OUT1 DR1, VCC2 +12V, OUT2 DR5. R1 22. Q1 IRF530 (G1). R4 100k. R3 2.2. SENSE_LOW.

R7 10k. DAMP. U2B ADG5412F: 14, S, D 15 GND. U2C ADG5412F: 11, S, D 10 GND. U2D ADG5412F: 6, S, D 7 GND.

16 IN SH. 8 IN RESET. 9 IN DP.

R9 0.05 / 0.5W. GND.

R8 10k +3V3. U2E ADG5412F: +12V 13 VDD, FF 12 FF; −12V 4 VSS, GND 5 GND.

J1 12V_INPUT_/_3V3_MONITOR: +12V_REG 1 +12V; −12V 2 −12V; 3V3_OUT 3 +3V3; GND 4 GND.

U4 INA240A1PW: IN+ 2, IN− 3, REF1 +3V3 7, REF2 GND 6; VS +3V3 5, OUT 8, GND 4 GND, NC 1. R10 100. CS_OUT. I_ADC. C2 10n GND.

J5 RAW_SHUNT_SENSE: CURRENT_SENSE+ 1 SENSE_LOW; CURRENT_SENSE− 2 GND.

J5 raw differential: 50 mV/A.
J3 I_ADC remains amplified.
Polarization current only.

LOCAL BYPASS: place each capacitor at its named IC supply pin

INA240: 1.65 V at zero current: 1 V/A.
J3–7 to PA0 / A0 (single-ended ADC).

DRAFT — component ratings / footprints still need review.
Current monitor measures polarization, not clamp current.
Kelvin route U4 input pair to R9 pads.
Signal and MCU labels join only the named remote nets.

C3 1u U1–8 +12V GND. C4 1u U1–6 +12V GND. C5 1u U3 +12V GND. C6 100n U2 +12V GND. C7 100n U2 −12V GND. C8 100n U4 +3V3 GND.

ECE 455 — Jun Yang

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Title: AWG19 Polarization — Connected Circuit

Size: A3	Date:	**Rev: DRAFT E**
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